

Requirement Specifications

Sentera 6X Thermal Pro Sensor with Gimbal For Freefly Astro Unmanned Aircraft System (UAS)

GENERAL INFORMATION

1.0 Scope of Work:

The USDA-ARS in Columbia, MO currently lacks an integrated payload for their Freefly Astro UAS that can simultaneously capture high-resolution red, green, blue (RGB), multispectral, and radiometric thermal data. This deficiency hinders efficient and consistent collection of crucial canopy-level data for crop stress detection, management decisions, and precision agriculture research. Acquiring a new Freefly Astro-compatible multi-sensor payload will enable synchronized imaging within a single flight. Consequently, this will lead to improved data quality, increased comparability, and better support for USDA-ARS research in crop monitoring and environmental stress assessment.

2.0 Background:

The USDA-ARS Cropping Systems and Water Quality Research Unit in Columbia, MO currently conducts research on long-term agricultural plots and fields, farmer fields, and research farms. This research requires high-resolution, multi-sensor aerial imagery to assess crop health, water stress, nutrient deficiency, and canopy temperature. Current research protocols depend on integrating visible red, green blue (RGB), multispectral, and thermal imagery collected simultaneously with an unmanned aircraft system (UAS) in order to assess crop responses to soil and agronomic treatments.

This USDA-ARS unit currently uses a Freefly Astro UAS platform with only an RGB camera. Purchasing single-sensor payloads will require multiple flights to capture the needed data and will require additional post-collection effort to standardize the different imagery layers onto the same geo-referenced area. Moreover, the ability to compare images will be hindered since they will be taken at different times in the day. The purchase of an integrated multi-sensor payload specifically designed for the Freefly Asto platform will eliminate these constraints. This capability is essential for maintaining continuity with multi-year datasets and supporting ongoing research into nutrient use efficiency, water use efficiency, tracking pests, and identifying management decisions.

CONTRACTOR REQUIREMENTS

3.0 Technical Requirements/Tasks:

Image Sensors

- 5 FPS Sustained Capture Rate
- Synchronized and Aligned Image Capture
- Four Multispectral Sensors
 - 3.2MP (Sony IMX265) IMX265Sony IMX265Sony
 - Global Shutter

- GSD at 200 ft / 60 m: 1.0 in / 2.6 cm
- 47° FOV
- Image Format - 12-bit TIFF
- Filter/Sensitivity
 - Green: 550nm center, 20 nm bandwidth
 - Red: 670 nm center, 30 nm bandwidth
 - Red Edge: 715 nm center, 10 nm bandwidth
 - NIR: 840 nm center, 20 nm bandwidth

Single RGB Sensor

- 20MP (Sony IMX147)
- Rolling Shutter
- GSD at 200 ft / 60 m: 0.4 in / 1 cm
- 47° FOV
- Image Format - 8-bit JPEG
- Filter/Sensitivity
 - IR Cut 650 nm

Single Radiometric Sensor

- FLIR Boson 640 x 512 LWIR
- Thermal Accuracy: +/- 5C
- GSD at 200 ft / 60 m: 3.4 in / 8.75 cm
- 50° FOV
- Image Format - 16-bit TIFF
- Filter/Sensitivity
 - 8 μm to 14 μm

Storage

- 512GB Internal SSD
- MicroSD (logging data)

Interfaces

- Freefly Smart Dovetail (on lightweight gimbal)
- USB-C
- Gigabit Ethernet

Dimensions (with with gimbal) 557.6

- Weight: 525g
- Add 58g if using included incident light sensor (ILS)

4.0 Government Furnished: Freefly-Astro UAV

5.0 Deliverables / Schedule:

All delivered to USDA-ARS-CSWQ at 1406 E. Rollins St.; 269 Ag Engineering Bldg.,
Univ. of Missouri, Columbia MO 65211

- One Sentera 6X Thermal Pro multi-sensor imaging payload (RGB + 4-band multispectral + FLIR Boson 640 thermal)
- One stabilized gimbal with Freefly Astro Smart Dovetail mounting interface
- One Integrated incident light sensor (downwelling light sensor) for multispectral radiometric calibration
- One factory-calibrated reflectance panel suitable for multispectral ground reference measurements Calibration

General description and platform compatibility

- Multi-sensor payload consisting of four 3.2 MP global-shutter multispectral imagers, one 20 MP RGB imager, and one FLIR Boson 640 radiometric thermal imager in a single gimballed unit.
- Must be fully compatible with the Freefly Astro UAS equipped with the Astro Isolator and use the Smart Dovetail quick-release interface for mechanical and electrical integration.
- Payload must include a stabilized gimbal designed for Freefly Astro that supports synchronized capture across RGB, multispectral, and thermal channels.

Optical and spectral performance

- RGB sensor must provide a minimum of 20 MP resolution with electronic rolling shutter.
- Multispectral sensors must provide four discrete bands (green, red, red-edge, near-infrared) with the following nominal center wavelengths and bandwidths or equivalent: green 550 nm $\pm$ 10 nm, red 670 nm $\pm$ 15 nm, red-edge 715 nm $\pm$ 5 nm, NIR 840 nm $\pm$ 10 nm.
- Thermal imager must be radiometric, long-wave infrared (LWIR) covering approximately 8–14 μ m with a minimum native resolution of 640 $\times$ 512 pixels.
- Horizontal field of view (HFOV) for RGB and multispectral channels must be approximately 47° to support standard crop-canopy mapping flight altitudes.

Radiometric and data-quality requirements

- System must provide radiometrically accurate thermal data suitable for pixel-level temperature analysis, including calibrated radiometric output from the Boson-class thermal core or equivalent.
- System must support radiometrically calibrated multispectral reflectance data using an incident light sensor and calibrated reflectance panel, included with the payload.
- All imaging channels (RGB, multispectral, thermal) must support synchronized and aligned image capture to enable multi-layer co-registered datasets from a single flight.

Acquisition rate, storage, and interfaces

- System must support a minimum sustained capture rate of 5 frames per second across imaging channels for efficient mapping missions.
- Payload must provide at least 512 GB of onboard solid-state storage for image data.
- Must support standard interfaces and protocols such as USB-C 3.0 and Gigabit Ethernet with PPS/serial/discrete I/O, and common UAS protocols including DJI/MAVLink or equivalent for command and data.
- Output formats must include 8-bit JPEG (RGB) and high-bit-depth TIFF for multispectral and thermal imagery (12-bit multispectral, 16-bit thermal or equivalent) for scientific analysis.

Physical, power, and operational requirements

- Total payload weight with Astro-compatible Smart Dovetail gimbal must not exceed approximately 525 g (excluding incidental accessories), to maintain Astro flight endurance.
- Power input must support 10.5–26 V range with typical power consumption near 15 W and maximum near 18 W.
- System must be NDAA-compliant for use in U.S. federal research programs.
- System must be field-deployable with accessories including: gimbal, incident light sensor, calibrated reflectance panel, mounting hardware, and required cables for integration with Freefly Astro.

Data and workflow

- Must provide full metadata tagging (time, GPS position, altitude, attitude, sensor settings) to support geospatial processing in common GIS and remote-sensing software.
- Must support “plug-and-play” installation on Freefly Astro with Smart Dovetail, allowing tool-less or minimal-tool mounting and removal in the field.
- Data products must be suitable for advanced agronomic analytics, including crop-stress mapping, canopy temperature mapping, and multi-temporal comparisons.